Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Scalable LLM inference system for lookup-augmented models			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		*Technology Field 2 is optional*	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	SoC		
Concept of Proposed Technology	DeepSeek and Meta have recently introduced LLMs that integrate static lookup modules [1, 2]. Both have shown that the lookup modules can be offloaded to CPU memory and prefetched to GPUs during inference, alleviating GPU memory pressure. While current models scale lookup modules to hundred-billion-parameter size, empirical results indicate that model quality continues to improve with larger lookup size. This trend suggests that next-generation LLMs will incorporate substantially larger lookup modules, potentially reaching trillion-parameter scale. At such scales, however, even offloading poses significant system-level challenges, as both CPU-side memory capacity requirements and CPU–GPU data movement overheads grow prohibitively large. To address these challenges, we propose a scalable LLM inference system for lookup-augmented models targeting data center deployments, utilizing emerging hardware technologies in conjunction with system-level software innovations. First, lookup modules are stored in a compressed form across a multi-tier memory hierarchy spanning DDR and local/shared CXL memory, with hotness-aware placement. Second, the system implements cooperative CPU–GPU computation, leveraging modern CPU AMX capabilities to perform computation around			

	lookup operations to reduce CPU–GPU data movement. Finally, lookup-aware request routing enables batched embedding transfers and efficient broadcasting in multi-GPU data-parallel inference, minimizing redundant communication.
Rationale for Joint R&D Project	The proposed project requires close coordination between system-level research and deployment-driven engineering throughout the development process. Key design decisions, including memory hierarchy design, execution partitioning between CPUs and GPUs, and request routing/clustering strategies for multi-GPU inference, must be evaluated not only for raw performance, but also for integration complexity, runtime robustness, and scalability. The Korean SME partner, such as Dnotitia, Rebellions, and Furiosa, brings a strong understanding of realistic deployment scenarios, including request arrival patterns, latency and throughput constraints, and software stack limitations encountered in production inference environments. This deployment insight informs our evaluation of architectural choices and runtime mechanisms, enabling selection of designs that are well suited for practical deployment, rather than idealized assumptions. In addition, assessment of end-to-end behavior beyond isolated benchmarking can be achieved through integration and testing of system components within commercially relevant inference frameworks and deployment infrastructures by the SME partner. Through this collaborative development process, continuous feedback between architectural exploration and system integration can be established, allowing design trade-offs to be identified early and refined through iterative prototyping and testing. This process is expected to improve the robustness of our design, reduce integration complexity, and ensure that research outcomes translate into deployable solutions aligned with production integration and operational constraints.
Keywords	1. Lookup-Augmented LLMs 2. Multi-Tier Memory Systems 3. CXL Memory Technology 4. CPU-GPU Heterogenous Computing 5. CPU On-Chip Accelerators

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (1)	End of the Project	Level (6-7)

R&D Objectives

Current Technical Capabilities – PI Nam Sung Kim's Lab (FAST Lab)

Professor Kim stands as the most prominent figure in computer architecture among Korean researchers in the field. He holds the distinct honor of being the first Korean scholar elevated to the prestigious status of both IEEE Fellow and ACM Fellow in his 40s. Furthermore, he is the first Korean researcher who became a member of the Hall of Fame (HoF) for all three premier computer architecture symposia: ISCA, MICRO, and HPCA. Beyond academia, his distinguished career includes a tenure as Senior Vice President at **Samsung's Memory Business Unit**, where he spearheaded the development of the world's first **HBM-based Processing-In-Memory (PIM)** architecture—a breakthrough that substantially enhances the performance and energy efficiency of emerging AI workloads. This seminal design currently serves as the cornerstone for commercial PIM products slated for market release in late 2026 or early 2027. Additionally, Professor Kim has catalyzed industry advancement by transferring many technologies to leading semiconductor titans through strategic patent licensing and open-source initiatives.

The FAST Lab has pioneered a suite of cutting-edge computing technologies, encompassing the optimization of state-of-the-art commercial systems for efficient execution of AI workloads, but also the design of general-purpose and domain-specific processors, near-memory processing units, and advanced system software. Transcending the boundaries of theoretical modeling and simulation typically done at academic research labs, the FAST Lab demonstrates its innovations through prototyping across ASIC, FPGA, and integration with commercial platforms. The FAST Lab epitomizes excellence in AI systems optimization, pioneering synergistic hardware-software co-designs that address critical bottlenecks in modern machine learning. Our research encompasses the development of high-throughput acceleration for billion-scale retrieval systems through advanced matrix instructions and the optimization of distributed training via in-network lossless compression [CABANA]. We possess a unique capability to actualize efficient LLM inference by harnessing on-chip accelerators for rapid decompression, orchestrating cooperative CPU-GPU computation through CXL offloading, and engineering TCO-efficient processing-near-memory platforms [LIA, CXL-PNM]. By bridging the gap between theoretical modeling and physical prototyping—including ASIC and FPGA implementations—the FAST Lab delivers transformative solutions that redefine the performance and energy efficiency of emerging AI workloads.

Overall Project Objectives

This project aims to develop a scalable large language model (LLM) inference system for lookup-augmented models targeting data center deployments, utilizing emerging hardware technologies in conjunction with system-level software optimizations. The proposed technology includes three complementary research directions:
(1) multi-tier memory management for cost-efficient scaling for CPU-side memory,
(2) CPU-GPU collaborative computation to minimize the CPU-GPU data transfer overhead, and
(3) lookup-aware request routing for multi-GPU inference systems to obviate redundant CPU-GPU communication.
Several technologies developed in our prior research provide a strong foundation for these directions and we plan to extend our technologies to address the requirements of lookup-augmented, large-scale inference workloads.

Multi-tier memory system

As DDR memory alone is often insufficient for hosting large lookup modules and large-capacity DIMMs (e.g., 128/256 GB) incur more than 2× higher per-GB cost, we propose a multi-tier CPU-side memory hierarchy for storing offloaded lookup embeddings. This hierarchy spans DDR memory as well as emerging memory technologies, including local CXL memory expanders and shared CXL memory pools. Although these technologies offer lower-cost capacity scaling and memory sharing across nodes, they exhibit higher access latency and lower bandwidth than DDR. Naively placing embeddings across these tiers can therefore significantly degrade inference latency and throughput.

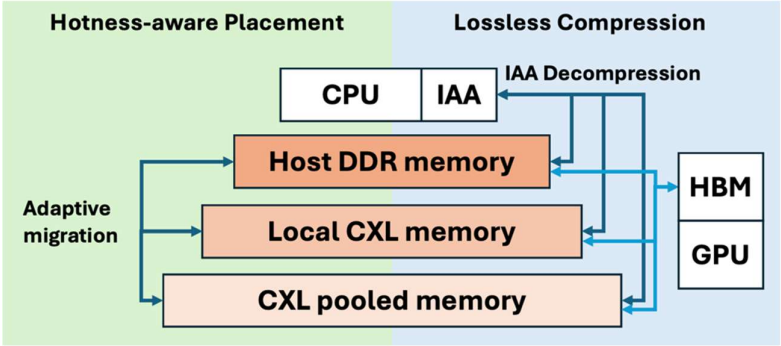

Figure 1. Proposed multi-tier memory system with hotness tracking and lossless compression

To address this challenge, we propose a hotness-aware placement mechanism for lookup embeddings. Natural language n-grams follow a Zipfian distribution [1], where a small subset of embeddings is accessed disproportionately frequently. Frequently accessed (hot) embeddings can thus be placed in higher tiers, while infrequently accessed (cold) embeddings reside in lower tiers. Moreover, embedding access patterns vary dynamically with workload context (e.g., code, mathematics, or natural language), requiring adaptive placement strategies.
In prior work, we developed M5 [3], a CXL-based page hotness tracking mechanism that enables efficient migration of frequently accessed pages from CXL memory to DDR, achieving a 14% improvement in tracking efficiency over CPU-based approaches. In this project, we will extend M5 by (i) transitioning from page-level to embedding-level access tracking, (ii) incorporating workload- and content-category-aware hotness estimation, and (iii) coordinating placement decisions across DDR, local CXL, and shared CXL memory pools.
To further reduce memory pressure, we will store lookup embeddings in a losslessly compressed form and decompress them on demand using Intel's on-chip In-Memory Analytics Accelerator (IAA). Our prior work LILo [4] developed a lossless compression framework for LLM inference, reducing BF16 parameter memory footprint by up to 33% while sustaining 150 GB/s decompression throughput. In this project, we will extend LILo to operate in a multi-tier memory setting by enabling compression across tiers, supporting concurrent decompression for batched requests, and integrating decompression scheduling with execution pipelines to minimize runtime overhead.

Collaborative CPU-GPU computation during inference.
Recent advances in Intel Advanced Matrix Extensions (AMX) have significantly increased CPU compute capability for LLM workloads. While DeepSeek and Meta offload only lookup storage and retrieval to the CPU for lookup-augmented LLM inference, additional performance gains are feasible by offloading subsequent linear operations to the CPU's AMX engine. By transferring only the computed outputs to the GPU rather than the raw embedding, CPU–GPU data movement can be reduced, resulting in lower inference latency and higher throughput. However, determining the appropriate extent of such offloading requires careful analysis of operation-level compute intensity, CPU and GPU throughput, memory bandwidth, and CPU–GPU interconnect bandwidth. In addition, overlapping AMX computation, GPU execution, and

data transfers requires careful orchestration of execution streams and synchronization granularity.
In our prior work, LIA [5], we developed an analytical model for determining optimal computation partitioning between GPUs and CPU AMX engines during LLM inference, achieving up to 3.7× throughput improvement by reducing CPU–GPU data transfer under CPU-side data offloading. LIA also introduced coordinated scheduling of CPU and GPU compute streams together with CPU–GPU data transfers to maximize overall resource utilization. In this project, we will extend LIA by adapting its analytical model to lookup-augmented LLMs, incorporating memory-tier access costs, and jointly optimizing computation placement across CPU AMX engines and GPUs. The resulting solution will be integrated into the PyTorch software stack to enable practical deployment.

Lookup-aware request routing for multi-GPU parallel inference.
In datacenter deployment scenarios, inference serving commonly utilizes multiple GPUs within a node. In such configurations, each GPU maintains a replica of the backbone LLM, while the lookup module is shared and offloaded to CPU memory. During inference, each GPU processes its own batch of requests and retrieves the required lookup embeddings from the CPU.
Without careful request routing, redundant embedding transfers occur. For example, if requests with similar token distributions are assigned to different GPUs, the same lookup embeddings may be transferred multiple times over the CPU–GPU interconnect. For large lookup modules, this redundancy can significantly exacerbate data transfer bottlenecks.
To mitigate this issue, we will develop a lookup-aware request routing mechanism that clusters requests with similar token distributions onto the same GPU, thereby reducing redundant embedding transfers. Furthermore, when lookup embeddings required by one GPU have already been fetched by another, peer-to-peer transfers over high-bandwidth interconnects such as NVLink will be prioritized rather than fetching the data again from the CPU. This effort will require new runtime and library support compatible with the PyTorch and HuggingFace Transformers software stack.

R&D Target Specifications

We target measurable improvements in inference performance (latency/throughput-related metrics) and deployment cost for lookup-augmented LLM inference in data center environments. All targets are defined relative to a baseline system that offloads lookup embedding storage and basic lookup operations to CPU-side DDR memory, without CPU–GPU collaborative computation or lookup-aware request routing. Our proposed system components combined, we aim to deliver a lookup-augmented LLM inference system for lower cost and improved inference latency as depicted in Figure 2.

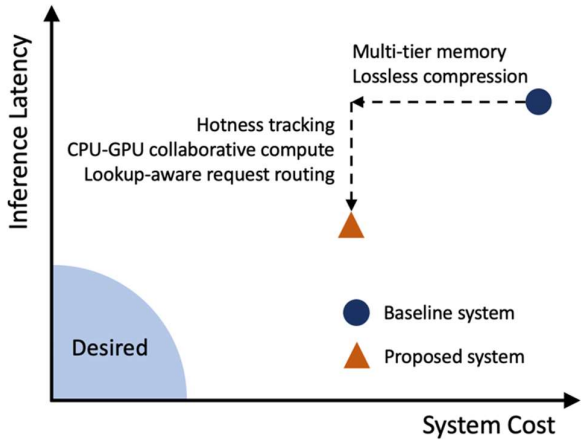

Figure 2. Inference latency and system cost reduction via proposed approaches

Inference performance improvement (latency and throughput)
First, the project targets a 30 percent improvement in inference performance across latency- and throughput-related metrics. Latency improvements will be measured using Time-To-First-Token (TTFT), defined as the time from request arrival to generation of the first output token, and Time-Per-Output-Token (TPOT), defined as the average latency per generated token during decoding. Throughput improvements will be measured using Requests Per Second (RPS) under steady-state serving conditions. These gains will be achieved through coordinated CPU–GPU collaborative computation, lookup-aware request routing in multi-GPU settings. The target improvement reflects end-to-end system behavior, including data transfer, computation overlap, and request batching, rather than isolated kernel-level speedups.

CPU-side memory cost reduction with bounded performance overhead
Second, the project targets a 50 percent reduction in CPU-side memory cost compared to a DDR-only baseline. This reduction will be achieved by leveraging a multi-tier memory hierarchy composed of DDR memory, local CXL memory expanders, and shared CXL memory pools, combined with lossless compression of lookup embeddings. The compression mechanism will reduce the effective memory footprint of lookup embeddings to 70 percent of the original BF16 storage size. Despite the use of higher-latency memory technologies and compression, the project targets an embedding retrieval overhead of no more than 5 percent relative to the DDR-only baseline, ensuring that memory cost savings do not significantly degrade inference latency or throughput.

R&D Plan

The R&D plan is organized into four sequential phases that align with the project's technical objectives as summarized in Figure 3. Each phase combines system design, prototype development, and validation, with close coordination between the research institute and the Korean SME partner.

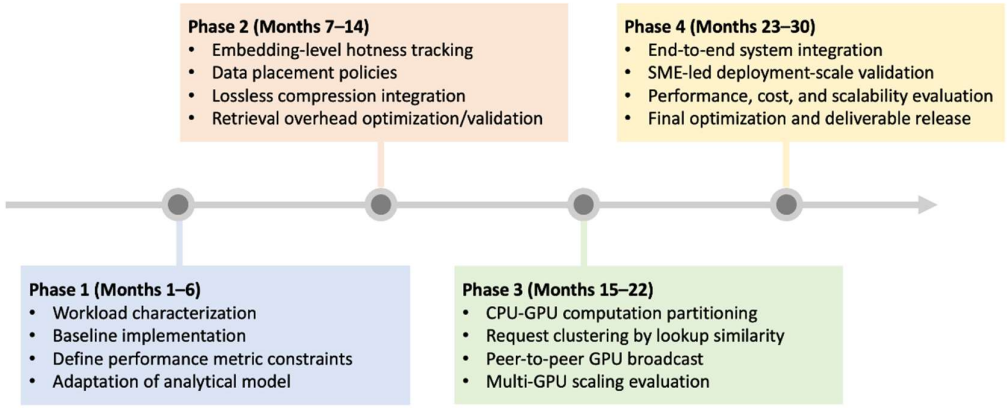

Figure 3. Project timeline showing phased R&D activities

Phase 1: System design and baseline characterization (months 1–6)

The project begins with defining system requirements and establishing baseline performance. During this phase, we will characterize lookup-augmented LLM inference characteristics, the amount of retrieved lookup embeddings, lookup access distribution across different inputs, compute characteristics around the lookup, etc. Baseline systems will be implemented based on CPU-side lookup storage and retrieval without collaborative computation or lookup-aware routing. These baselines will be used to quantify memory usage, CPU–GPU data movement, TTFT, TPOT, and throughput, serving as reference points for later evaluation. The analytical models from prior work will be reviewed and adapted to incorporate lookup-augmented inference characteristics.

Milestone: Validated baseline system and workload characterization report.

Phase 2: Multi-tier memory management and compression integration (months 7–14)

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In this phase, we will develop and integrate the multi-tier CPU-side memory system for lookup embeddings. This includes extending hotness tracking from page-level to embedding-level granularity, incorporating workload- and context-aware access modeling, and implementing dynamic placement and migration policies across DDR, local CXL memory, and shared CXL memory pools. Lossless compression will be integrated using the existing LILo framework, with extensions to support compression across memory tiers and concurrent decompression for batched inference requests. Performance overhead introduced by compression and tiered access will be carefully measured and optimized.

Milestone: Functional multi-tier memory system with hotness-aware placement and bounded retrieval overhead.

Phase 3: CPU–GPU collaborative inference and request routing (months 15–22)

This phase focuses on reducing CPU–GPU data movement and improving inference performance. We will extend analytical models for computation partitioning to lookup-augmented LLMs, incorporating memory-tier access costs and interconnect bandwidth constraints. Based on these models, collaborative CPU–GPU execution mechanisms will be implemented, leveraging CPU AMX engines for selected linear operations. In parallel, lookup-aware request routing and clustering mechanisms will be developed for multi-GPU inference to reduce redundant embedding transfers and exploit peer-to-peer GPU interconnects when available. All mechanisms will be integrated into the PyTorch software stack.

Milestone: Integrated CPU–GPU collaborative execution and lookup-aware request routing prototype.

Phase 4: System integration, validation, and optimization (months 23–30)

In the final phase, the complete system will be integrated and evaluated under realistic deployment scenarios provided by the Korean SME partner. End-to-end inference performance, memory cost reduction, scalability, and robustness will be evaluated against the target specifications. Iterative optimization will be performed to address integration bottlenecks and performance regressions. The project will conclude with a comprehensive evaluation report and a deployable system prototype suitable for technology transfer and commercialization.

Milestone: Final system meeting target specifications with validated deployment readiness.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	We will lead the system-level research and architectural design of the proposed lookup-augmented LLM inference system, with primary responsibility for developing, prototyping, and evaluating novel mechanisms that extend beyond current production capabilities. Our technical scope covers multi-tier memory systems, heterogeneous CPU–GPU execution, and runtime-level optimizations for large-scale inference workloads. Specifically, we will design and implement hotness-aware memory management mechanisms for lookup embeddings across DDR memory, local CXL memory expanders, and shared CXL memory pools. This includes extending prior CXL-based hotness tracking techniques to embedding-level granularity, integrating workload- and context-aware access modeling, and developing placement and migration policies tailored to lookup-augmented inference. We will also develop CPU–GPU collaborative execution mechanisms that leverage CPU on-chip accelerators, such as AMX and IAA, to reduce CPU–GPU data movement and improve inference performance. This includes extending analytical models for computation partitioning in lookup-augmented models, implementing coordinated scheduling of CPU and GPU compute and data transfers, and integrating these mechanisms into the PyTorch software stack. In addition, we will develop lookup-aware request routing and batching strategies for multi-GPU inference and conduct rigorous performance evaluation using controlled micro-benchmarks and end-to-end inference workloads. we will provide access to advanced experimental platforms, including multi-GPU servers and CXL-enabled systems, as well as expertise in system instrumentation, performance analysis, and prototype development. We will collaborate closely with the Korean SME partner to align research outputs with deployment constraints and to support integration and validation efforts.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to lead the deployment-driven engineering, system integration, and validation activities necessary to translate research outcomes into deployable and commercially viable inference systems. The SME's primary role is to provide practical deployment context, implement research prototypes within production-oriented software stacks, and validate system behavior under realistic operating conditions. Specifically, the SME partner will define representative inference workloads, including request arrival characteristics, latency and throughput constraints, and operational requirements relevant to data center deployment. These inputs will guide architectural and runtime design decisions throughout the project. The SME partner will also be responsible for integrating research-developed technologies, such as multi-tier memory management, compressed lookup storage, CPU–GPU collaborative execution, and lookup-aware request routing, into existing inference frameworks such as vLLM and their serving infrastructures. In addition, the SME partner will conduct system-level testing and validation in commercially relevant environments. This includes evaluating end-to-end inference performance, stability, and scalability under realistic load conditions, and identifying integration challenges that may not surface in controlled research experiments. The SME will commit engineering personnel, development infrastructure, and access to representative datasets and deployment configurations to support iterative co-development and validation. Through close technical collaboration, regular design reviews, and shared evaluation efforts, the SME partner is expected to play a role in ensuring that the proposed technologies mature beyond research prototypes and are suitable for real-world deployment and commercialization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver a deployable, scalable inference system for lookup-augmented LLMs that addresses both performance and cost challenges in data center environments. From a technological perspective, the project will produce validated system architectures and software prototypes that integrate multi-tier memory management across DDR and CXL-based memory, lossless compression with bounded overhead, cooperative CPU–GPU execution leveraging on-chip accelerators, and lookup-aware request routing for multi-GPU inference. These outcomes will demonstrate measurable improvements in inference latency, throughput, and memory cost efficiency under realistic deployment conditions. From a commercialization standpoint, the project will enable the Korean SME partner to incorporate advanced inference system capabilities into production-ready platforms and services. The resulting technologies are expected to reduce infrastructure costs, improve serving efficiency, and increase scalability for lookup-augmented LLM applications, thereby accelerating time-to-market and lowering technical risk for commercial deployment. The joint development and validation process will further enhance the SME's system integration expertise and competitiveness in global AI infrastructure markets. At a broader level, the project contributes to strengthening the AI systems ecosystem by advancing practical use of emerging hardware technologies such as CXL memory and CPU on-chip accelerators. The outcomes are expected to promote wider adoption of cost-efficient, large-scale AI inference solutions, support workforce development in advanced system engineering, and foster sustained international collaboration with positive economic and industrial impact.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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